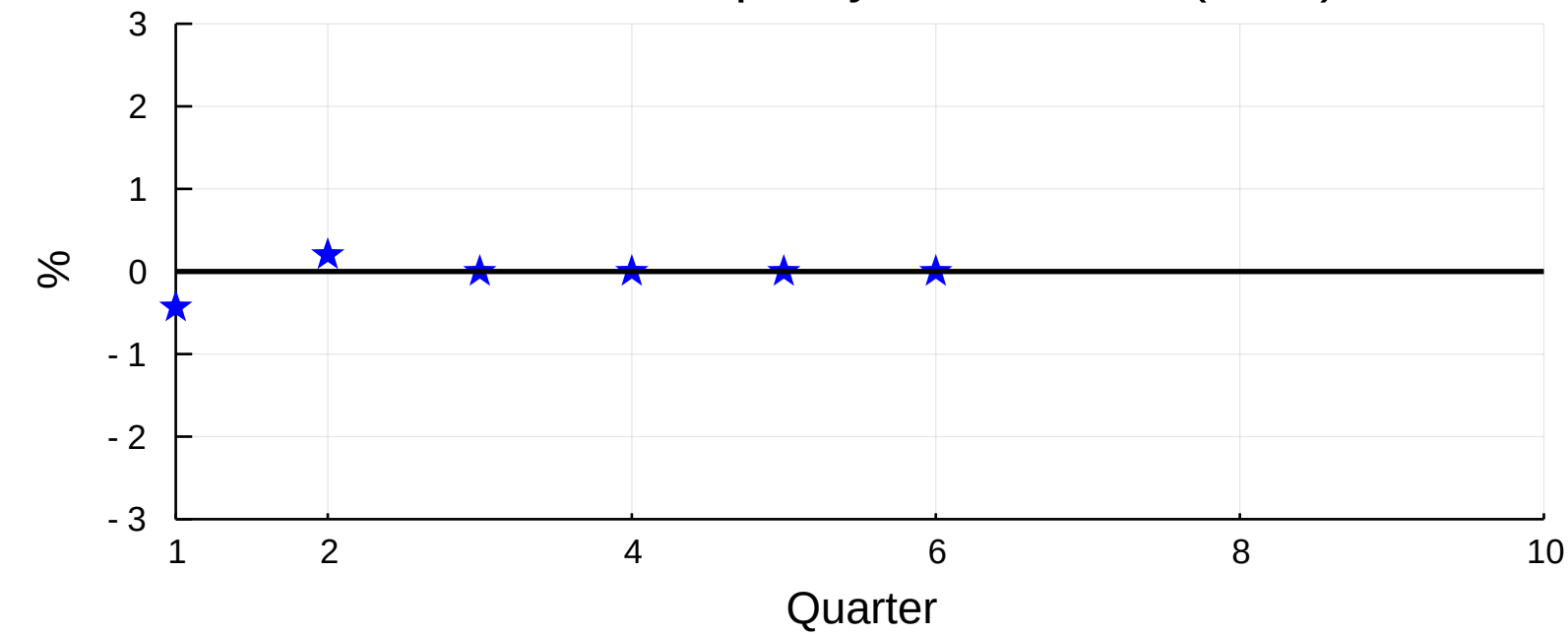
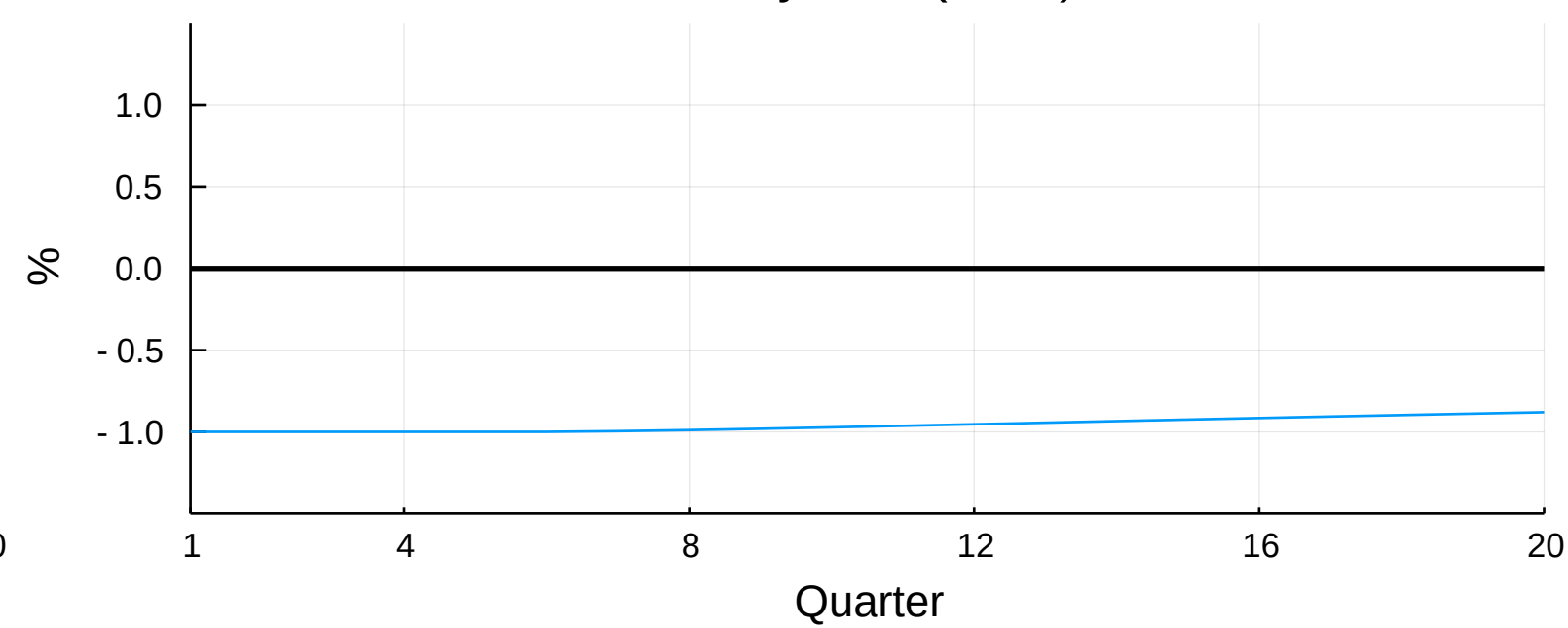


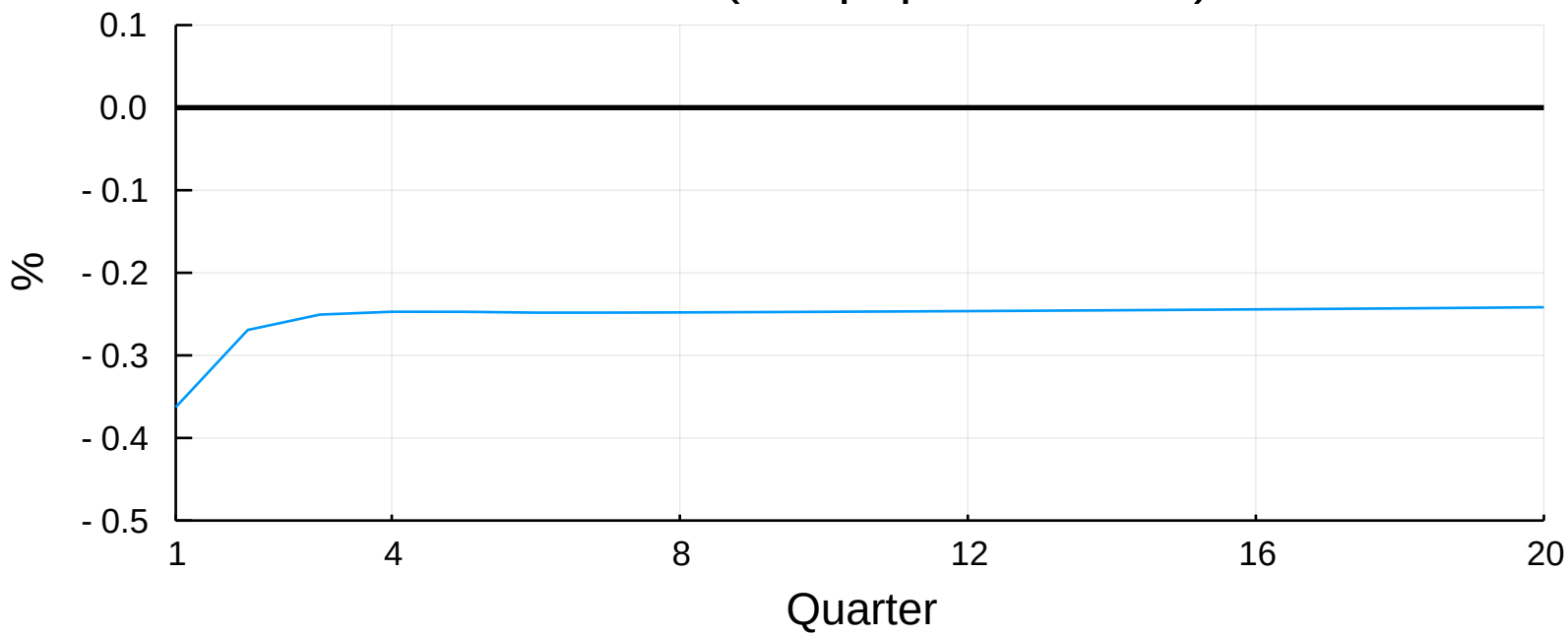
Permanent Liquidity Innovations (APR)



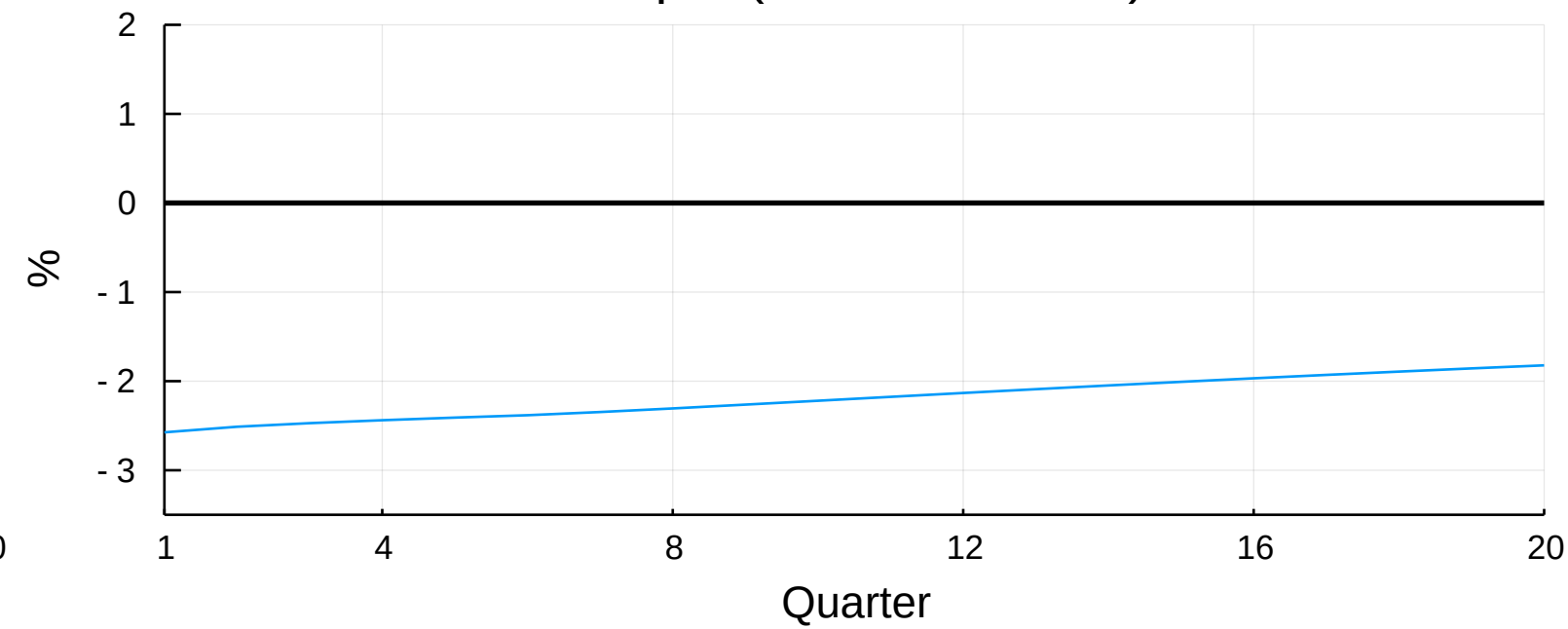
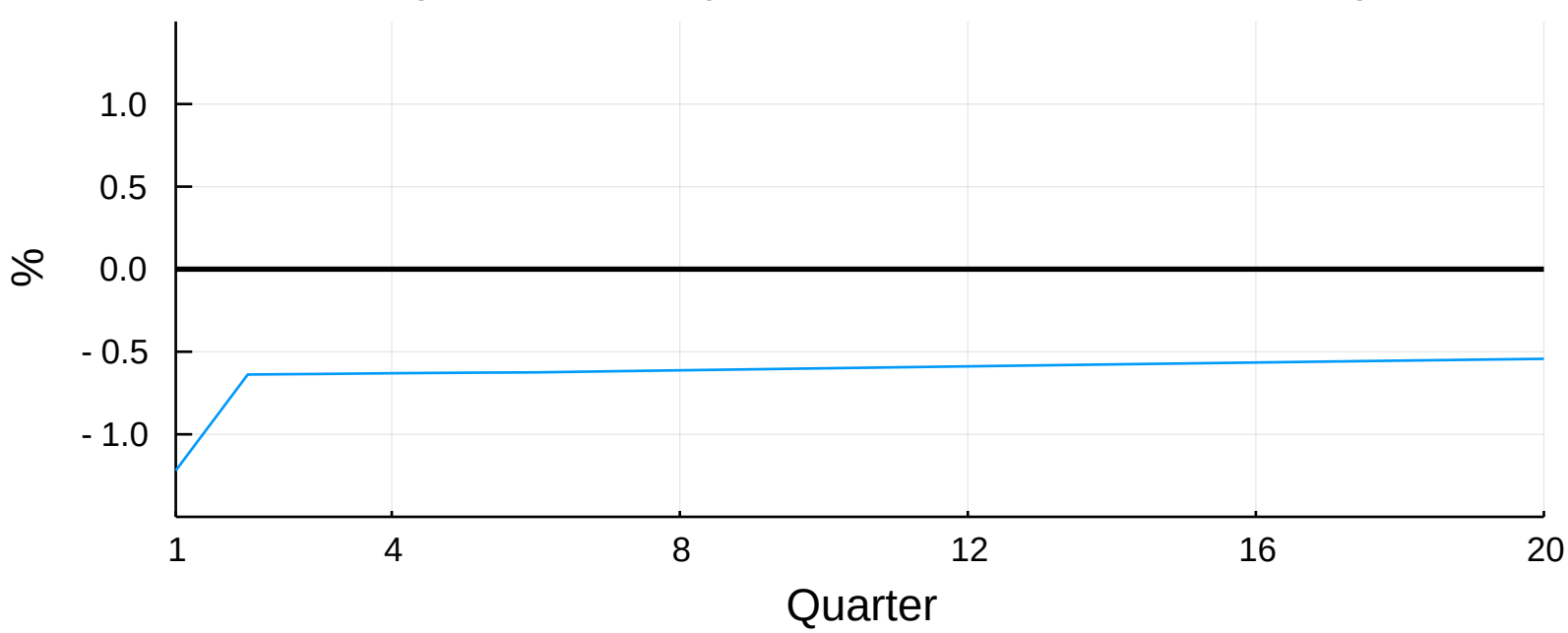
Policy rate (APR)



Inflation (% , qoq annualized)



Output (% dev from SS)

 r^* (Forward 5- year real natural rate, APR)

Ex- ante real rate (APR)

